

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P1-Model)** **Date: 08-02-2026**

Time: 09.00Am to 12.00Pm **GTA-15** **Max. Marks: 180**

08-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2023-P1)_GTA-15_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2023-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				**17**	**60**

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CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

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MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

1. Let $A\left(t_1^2,2t_1\right),B\left(t_2^2,2t_2\right),C\left(t_3^2,2t_3\right),(t_i\neq 0\ \forall i=1,2,3)$ be three distinct points on the parabola $y^2=4x$. If orthocenter of ΔABC is the focus $S(1,0)$ of the parabola $y^2=4x$, then

A) $t_1t_2+t_2t_3+t_3t_1=-5$ **B)** $t_1+t_2+t_3+t_1t_2t_3=0$

C) $\frac{1}{t_1t_2}+\frac{1}{t_2t_3}+\frac{1}{t_3t_1}=-1$ **D)** $(1+t_1)(1+t_2)(1+t_3)=-4$

2. S is a focus of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. P and Q are two points on the ellipse such that for all positions of the point P on the ellipse, perimeter of the ΔPQS is always constant. The maximum area of ΔPQS is (where e is eccentricity of ellipse)

A) $2\sqrt{2}\ b^2e\ \forall\ e\in\left(0,1/\sqrt{2}\right)$ **B)** $\sqrt{2}\ ab\ \forall\ e\in[1/\sqrt{2},1)$

C) $2\sqrt{2}\ b^2e\ \forall\ e\in[1/\sqrt{2},1)$ **D)** $\sqrt{2}\ ab\ \forall\ e\in\left(0,1/\sqrt{2}\right)$

3. The coefficients of the quadratic function f(x) including the constant term, are all rational has local maximum at x = 0. Let $g(x)=\left|f'(x)\right|e^{f(x)}$ has maximum value $4\sqrt{e}$. If $g(x)=4\sqrt{e}$ has rational solutions then

A) $\int_{-1}^{0}g(x)dx=e-\frac{1}{e^7}$ **B)** The value of $\operatorname{sgn}\left(f(0)\right)=-1$

C) g(x) is non derivable at one value of x **D)** The value of $g\left(\tan\frac{\pi}{4}\right)=\frac{2}{e^7}$

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SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

4. How many four 4 digit numbers are possible such that during addition there is no carry over at any stage ?

A) $\left({}^{13}C_4\right)^3$ **B)** $\left({}^{9}C_4\right)^3\ {}^{13}C_4$ **C)** $\left({}^{13}C_4\right)^3\ {}^{9}C_4$ **D)** $\left({}^{13}C_3\right)^3\ {}^{9}C_3$

5. $f:\mathbb{R}\to\mathbb{R}$ is a continuous and differentiable function satisfying

$$e^x+e^{-x}\int_0^x f(t)\,dt-1=\int_0^x \frac{\int_0^t e^x f(x)\,dx}{e^t}\,dt;\ f(0)=3,\text{then } f(1)=?$$

A) $2e+1$ **B)** $2e+e^e$ **C)** $e+2e^e$ **D)** $2+e^e$

6. Let $g(k,z)=Z^k\int_0^{1/z} t^{t+k-1}dt$ if $\lim_{z\to\infty} g(k,z)=f(k)$ then no. point of interaction of curves $xy=\dfrac{1}{f(x)}$ and $x^2+y^2=8$ is.

A) 0 **B)** 2 **C)** 4 **D)** 8

7. $I_1=\int_0^1\left(1-\left(1-x^3\right)^{\sqrt{2}}\right)^{\sqrt{3}}x^2\,dx$, $I_2=\int_0^1\left(1-\left(1-x^3\right)^{\sqrt{2}}\right)^{\sqrt{3}+1}x^2\,dx$ then $\dfrac{I_1}{I_2}-1$ is equal to :

A) $\dfrac{1}{\sqrt{2}\left(\sqrt{3}-1\right)}$ **B)** $\dfrac{\sqrt{3}-1}{2\sqrt{2}}$ **C)** $\dfrac{\sqrt{3}}{2\sqrt{2}}$ **D)** $\dfrac{\sqrt{3}+1}{2\sqrt{2}}$

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. $f:\mathbb{R}\to\mathbb{R}$, $f(x)$ is a continuous and differentiable function such that

$f(x)=e^x\left(f'(x)+f''(x)+f'''(x)+.....\infty\right)$ and $f(0)=\frac{1}{4}$, then the value of $\lim\limits_{x\to\infty}f(x)$ is :

9. $\lim\limits_{t\to\infty}e^{4x}t\int\limits_0^1 x^t g(x)\,dx\le f''(x)-4f'(x)+3f(x)\le\lim\limits_{t\to\infty}2e^{5x}t\int\limits_0^1 x^t h(x)\,dx$ where $\frac{f(x)}{e^x}$ is an increasing function of 'x'. The area bounded by $y=f(x)$ and $y=f'(x)$ between $x=0$ and $x=\ln 2$, where $f(0)=f'(0)=0$ and $g(x)=\frac{11x^2+x}{13x+4}$, $h(x)=\frac{13x+2}{11x^2+47x}$ in sq. units is ?

10. $f:\mathbb{R}\to\mathbb{R}$ is a continuous and differentiable function such that

$\frac{f(x^2)}{1+e^x}+\frac{f'(x^2)}{1-e^x}+x^2=\int\limits_{\int_0^x f(t^2)dt}^{\int_0^x f(t^2)dt} f(t)\,dt$; $f(1)=\frac{1}{e}$, then value of $f(0)$ is

11. Let $I_n=\int\limits_0^1 x^n\left(\cot^{-1}x\right)dx$ and $J_n=\int\limits_0^1 x^n\left(\tan^{-1}x\right)dx$.

Then $13I_{12}+11I_{10}+10J_9+8J_7=\pi-\frac{m}{n}$, where m and n are coprime then value of $\left[\frac{n}{m}\right]$.

[.] = greatest integer function.

12. A drunken man stands with a cliff one step to his left. He takes steps randomly left and right. Each step has probability $\frac{1}{6}$ of going left and probability $\frac{5}{6}$ of going right. Each step is the same size. If allowed to randomly step indefinitely, let 'p' be the probability that the drunken man escape the cliff, then the value of 5p is ?

13. Let points A $(x_1,y_1,0)$ and B $(x_2,y_2,0)$ are lying on curve $y=2+e^{\ln(1-x)}+\log_{1/2}2^{\sqrt{x^2-2x+1}}$, and A also lies an curve $x^2+y^2=13, z=0$ and B lies inside it. If x_1,y_1,x_2,y_2 are integers, for largest value of x_2 shortest distance of A from line $\frac{x-x_1}{x_2+3}=\frac{y-y_2}{4}=\frac{z-3}{5}$ is $\frac{a}{\sqrt{b}}$, a and b are coprime, then find the value of $a+b$.

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SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14.

	List – I		List – II
(I)	Let $S = {}^{31}C_1 - \left(1+\frac{1}{2}\right){}^{31}C_2 - \left(1+\frac{1}{2}+\frac{1}{3}\right){}^{31}C_3 - \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right){}^{31}C_4 + \ldots$ $\ldots + \left(1+\frac{1}{2}+\frac{1}{3}+\ldots\ldots\ldots\ldots+\frac{1}{31}\right){}^{31}C_{31}$, then $\frac{1}{S}$ is equal to	(P)	16
(II)	If $\frac{{}^9C_0}{8} - \frac{{}^9C_1}{9} + \frac{{}^9C_2}{10} - \frac{{}^9C_3}{11} + \ldots\ldots\ldots + \frac{{}^9C_8}{16} - \frac{{}^9C_9}{17} = \frac{1}{n}$ then n is divisible by	(Q)	27
(III)	Remainder when $43^{43^{43}}$ is divided by 40 is	(R)	13
(IV)	Let $S_n = \sum_{k=0}^{n} \frac{{}^{n+k}C_k}{2^k}$, then remainder when S_{12} is divided by 40	(S)	31
		(T)	17

A) $I \to R; II \to R,S; III \to Q; IV \to P$ **B)** $I \to S; II \to R,T; III \to Q; IV \to P$

C) $I \to R; II \to R,Q; III \to S; IV \to S$ **D)** $I \to S; II \to R,T; III \to Q; IV \to S$

15. Let A {1, 2, 3, 4}, B {1, 2, 3, 4, 5, 6, 7, 8} then match the list I & II.

	List – I		List – II
(I)	Then number of function $f: B \to A$, such that if $x_1 > x_2$, then $f(x_1) \geq f(x_2) \forall x_1 \in B$ is N. Then the sum of digits of N is	(P)	2
(II)	Then number of onto function $f: B \to A$, such that if $x_1 > x_2$, then $f(x_1) \leq f(x_2) \forall x_1 \in B$ is N. Then the sum of digits of N is	(Q)	8
(III)	The number of one-one function $f: A \to B$ such that $f(x) \neq x \forall x \in A$ is N. Then the sum of digits of N is	(R)	9
(IV)	The number of one-one function $f: A \to B$ such that there are exactly two, x, $x \in A$ for which $f(x) = x$ is N. Then the sum of digits of N is	(S)	7
		(T)	12

A) $I \to Q; II \to T; III \to R; IV \to P$ **B)** $I \to Q; II \to T; III \to P; IV \to R$

C) $I \to T; II \to Q; III \to P; IV \to R$ **D)** $I \to T; II \to Q; III \to S; IV \to R$

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16.

	List – I		List – II
(I)	Let $f(x)$ be a non negative continuous function $\forall x \geq 0, f(0)=0$. If $f'(x) \leq f(x)\sin x \ \forall x \geq 0$, then the value of $f\left(\frac{3\pi}{2}\right)$ is	(P)	4
(II)	If $a\left(x^2+1\right)+12x+6-3x^3 < \sin^{-1}\sin(100)+\cos^{-1}\cos(100)$. $\forall x \in R, a \in I$, then "a" can not be equal to	(Q)	3
(III)	If two events A and B are such that $P(A^C)=0.35, P(B)=0.4$ and $P(A \cap B)=0.15$, then $20P\left(\frac{B}{A \cup B^C}\right)$ is	(R)	0
(IV)	The number of elements in the range of the function $f(x)=\sqrt{\cos^{-1}\left(4x^3-3x\right)-\frac{\pi}{2}}+\frac{(2x+3)!}{\sqrt{x+2}}$ is	(S)	5
		(T)	8

A) $I \to P; II \to P,Q; III \to R; IV \to T$ **B)** $I \to R; II \to P,Q,R; III \to P; IV \to T$

C) $I \to P; II \to Q,R; III \to S; IV \to Q$ **D)** $I \to R; II \to P,Q,R,S; III \to P; IV \to Q$

17. Match the List – I with List – II.

	List – I		List – II
(I)	The shortest distance between the curves $2y^2=x^3$ and $9x^2+9y^2-60y+91=0$ is $\frac{a\sqrt{13}-b}{3}$, a, b are rational number then $a+b$ is	(P)	5
(II)	The maximum value of the expression $\tan\left(x+\frac{2\pi}{3}\right)-\tan\left(x+\frac{\pi}{6}\right)+\cos\left(x+\frac{\pi}{6}\right)$ in the interval $\left[-\frac{5\pi}{12},-\frac{\pi}{3}\right]$ is $\frac{11\sqrt{3}}{k}$ then k is	(Q)	4
(III)	Let $L=\lim_{x\to\infty} x\left(2x-\sqrt[3]{x^3+x^2+1}-\sqrt[3]{x^3-x^2+1}\right)$ then $9L$ is equal to	(R)	2
(IV)	Let $f(x)$ be a differentiable function with $f'(0)=2$ and $f(0)=3$ such that $f'(x)=\frac{f(x+n)-f(x)}{n} \forall x \in R$ and $n \in N$, then $\frac{f(6)}{f(1)}$ is	(S)	3
		(T)	6

A) $I \to P; II \to T; III \to S; IV \to R$ **B)** $I \to P; II \to T; III \to R; IV \to S$

C) $I \to T; II \to P; III \to R; IV \to S$ **D)** $I \to P; II \to T; III \to S; IV \to Q$

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PHYSICS **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

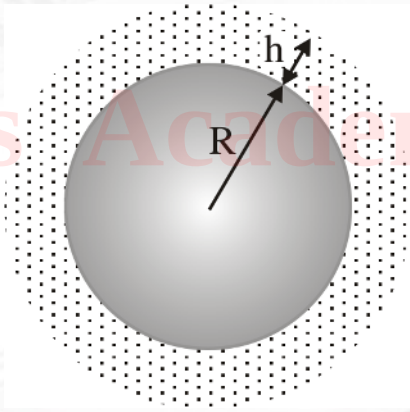
This section contains **THERE (03)** questions.

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Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

18. During fair weather conditions atmosphere of the Earth can be considered as a good conductor with resistivity ρ depending on the height h from the surface of the Earth as $\frac{1}{\rho} = a + bh^2$, where a and b are constants. It is also known that near the surface of the Earth is a downward electric field E_0, which causes a radial current in the atmosphere. Assume E_0 is constant in magnitude.



A) The current density J decreases with increase in height.

B) The electrons will be travelling in radially outward direction.

C) The charge density will be negative at any general height.

D) The charge density at height h is $\frac{2\varepsilon_0 E_0}{R+h}$ in in magnitude.

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19. Unpolarized light is incident on a prism of angle 45° and refractive index 4/3. The ray partially reflected at the first surface is completely polarized. Choose the correct statement(s).

A) The angle of deviation is approximately 18.7°.

B) The transmitted ray at the second face is completely polarized.

C) The transmitted ray at the first face is completely polarized.

D) There is no partial reflection at the second face.

20. A car of mass m is moving on a horizontal surface with speed v along positive x-axis. The driver of car wishes to accelerate the car. He accelerates the car which provides a constant tractive force F, directed along the line of motion of car. The frontal force of resistance is directed opposite to the velocity and given as F = αv in magnitudes where α is a positive constant and v is speed of car at that time. Assume initially car is at origin. Express the tractive force F as a function of x and v, where x is the displacement of the car and v is the velocity at that position.

A) $\alpha\dfrac{V^2-V_0^2e^{\frac{-2\alpha x}{m}}}{1-e^{\frac{-2\alpha x}{m}}}$

B) $\alpha\dfrac{V^2+V_0^2e^{\frac{-2\alpha x}{m}}}{1-e^{\frac{-2\alpha x}{m}}}$

C) $\alpha\dfrac{V^2-V_0^2e^{\frac{-2\alpha x}{m}}}{1+e^{\frac{-2\alpha x}{m}}}$

D) $\alpha\dfrac{V^2+V_0^2e^{\frac{-2\alpha x}{m}}}{1+e^{\frac{-2\alpha x}{m}}}$

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

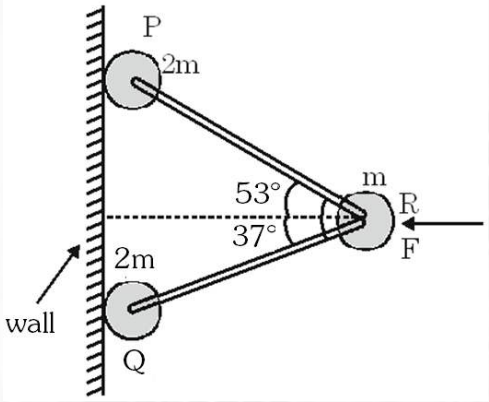
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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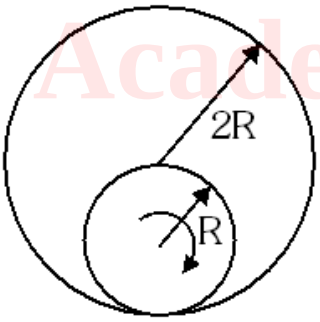
21. Three small discs are connected with two identical massless rods as shown in figure. The rods are pinned to the discs such that angle between them can change freely. The system is placed on a smooth horizontal surface with discs P and Q touching a smooth wall. A force F is applied to the disc R is in direction perpendicular to the wall. Find acceleration of disc P immediately after the force starts to act. Masses of discs are $m_P = m_Q = 2m$

[wall is perpendicular to the plane of the figure.]



A) $\frac{F}{\sqrt{2m}}$ **B)** $\frac{F}{8m}$ **C)** $\frac{2F}{11m}$ **D)** None of these

22. A wheel of radius R is rolling without sliding inside a fixed circular cylinder of radius 2R as shown. What is the trajectory followed by a point on the rim of the wheel ?



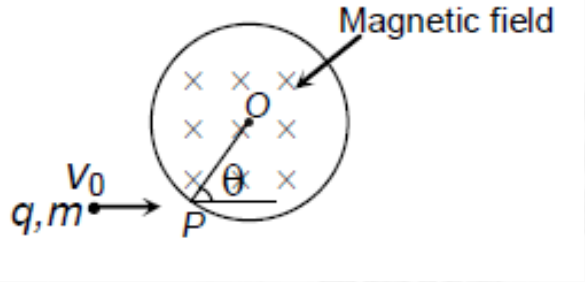
A) Circle **B)** Cycloid **C)** Straight Line **D)** None of these

23. There is a very small hole in a totally enclosed heated furnace. Outside the furnace, the air temperature is 0°C and the air pressure is 100kPa. The air inside the furnace is kept at a constant temperature of 57°C by the controlled heating system, and after a sufficiently long time its pressure also becomes stationary. Calculate (to the nearest integer) the magnitude of this stationary pressure.

A) 105kPa **B)** 120kPa **C)** 110kPa **D)** 124kPa

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24. A particle of charge q and mass m is projected with a velocity v_0 towards a circular region having uniform magnetic field B perpendicular and into the plane of paper from point P as shown in the figure. R is the radius and O is the centre of the circular region. If the line OP makes an angle θ with the direction of v_0 then the value of v_0 so that particle passes through O is



A) $\frac{qBR}{m\sin\theta}$ **B)** $\frac{qBR}{2m\sin\theta}$ **C)** $\frac{2qBR}{m\sin\theta}$ **D)** $\frac{3qBR}{2m\sin\theta}$

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

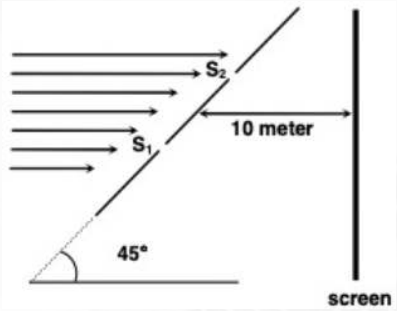
Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. A certain planet of radius R is composed of a uniform material that, through radioactive decay, generates a net power P such that heat per second $\frac{\Delta Q}{\Delta t} = P\frac{r^3}{R^3}$, where r is radial distance from centre. This results in a temperature differential between the inside and outside of the planet as heat is transferred from the interior to the surface. The rate of heat transfer inside is governed by conduction. It is found that thermal conductivity k is constant for the planet. For the following assume that the planet is in a steady state; temperature might depend on position, but does not depend on time. (Assuming black body radiation, and emissivity is 1) Find an expression for the temperature difference between the surface of the planet (T_s) and the centre of the planet (T_C). If your answer is

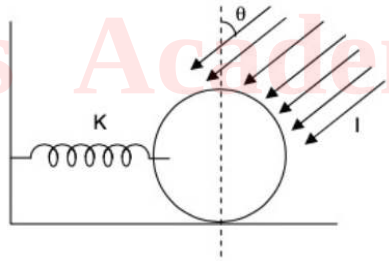
$\Delta T = \frac{\sigma R^m T_S^n}{\ell k}$ fill value of (m + n + ℓ)

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26. Consider a plane inclined at an angle 45° with the horizontal has two slits (S_1 and S_2) separated by a distance d = $\sqrt{2}$ mm. The screen is placed at a distance of D = 10 m. A parallel monochromatic light beam of wavelength $5000\,\mathring{A}$ is incident on the slits as shown. If the fringe width of interference pattern on the screen is $k\times10^{-3}$ meter, then find the value of k.



27. A sphere of radius R is kept on smooth horizontal surface. It is connected with ideal spring of constant k as shown in figure. A parallel light beam having intensity I is falling at an angle $\theta = 30^o$ with vertical on the sphere, then find the compression in the spring (in SI unit) in equilibrium position, assuming that the sphere absorbs 100% of the light incident on it. [Take $\frac{IR^2}{Kc} = \frac{10}{\pi}$ SI unit, where c is speed of light in vacuum]

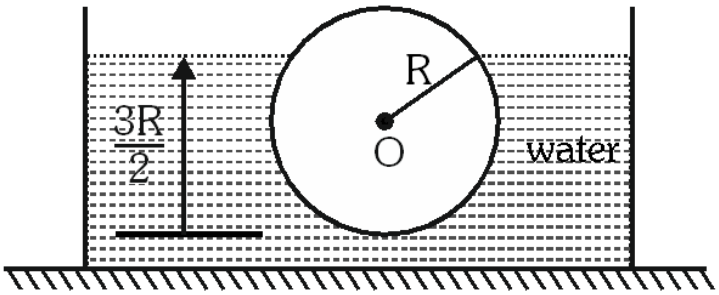


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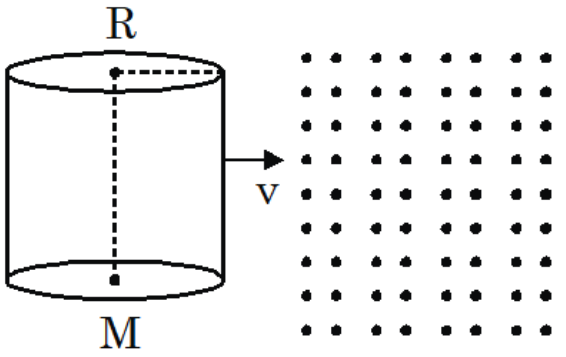
28. A vernier calliper has no zero error, one main scale division = 1 mm and 9 main scale division = 10 vernier scale divisions. A spring is held between jaws without exerting force on jaws. Main scale reading is 3.4 cm and 5th vernier scale division coincides with a main division. When Jaws are pressed by force of 10 N on both sides, main scale reading remains 3.4 cm, but 1st division of vernier scale coincides with a main scale division. Now spring is removed, cut into two pieces and when one part is inserted between jaws without any force, main scale reading = 1.1 cm and 5th vernier scale division coincides with a main scale. If we apply force of 30 N on both the jaws, main scale reading remains 1.1 cm and x vernier scale division coincides with a main scale division. Find x.

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29. A uniform solid sphere of radius R floats in water (density ρ_ω) such that distance between the water surface and lowest end of the sphere $\frac{3R}{2}$ is as shown in the figure. Then the specific gravity of the sphere is……



30. A cylinder of mass M and raidus R moves with constant speed v through a region of space that contains dust particles of mass m which is at rest. There are n number of particles per unit volume. The cylinder moves in a direction perpendicular to its axis. Assume m << M, and assume the particles do not interact with each other. All the collision takes place is perfectly elastic and the surface of the cylinder is smooth. The drag force per unit length of the cylinder require to maintain the speed v constant for the cylinder when it has entered in the region is $\frac{K}{3}nmRv^2$. Find the value of K.



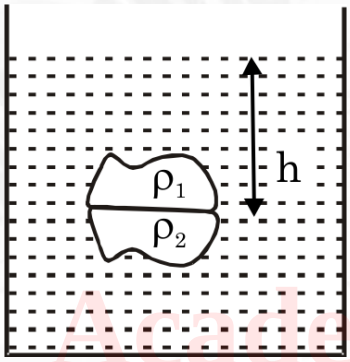
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SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

31. If objects shown in figure has volume $V_1 = 1m^3$ and $V_2 = 1m^3$ densities $\rho_1 = 400kg/m^3$ and $\rho_2 = 600kg/m^3$ and identical flat portion has area S = 100 cm^2 and if the glue can withstand a maximum force $F_{max} = 500N$ then :- (Neglect atmospheric pressure)



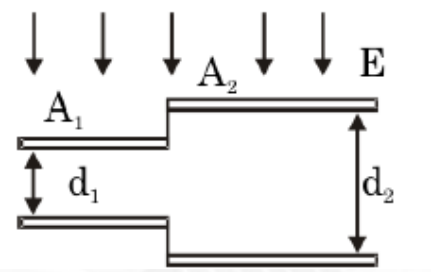
	COLUMN-I		COLUMN-II
I)	Value of h	**P)**	10 m
II)	Force by liquid on upper object	**Q)**	5500 N
III)	Force by liquid on lower object	**R)**	4500 N
IV)	Total force by liquid on both object	**S)**	10^4N
		T)	3000 N
		U)	7500 N

A) I–P, II–T, III–U, IV–S **B)** I–P, II–R, III–T, IV–S

C) I–P, II–R, III–Q, IV–S **D)** I–P, II–Q, III–R, IV–S

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32. A parallel plate capacitor has two sections. In these sections areas of plates are A_1 and A_2 and distances between the plates are d_1 and d_2 respectively. The distances between the plates are much smaller than their linear dimensions. An uniform electric field E that is perpendicular to the plates is switched on. Neglecting edge effects, E_1 and E_2 are electric fields established in sections (1) and (2) respectively.



List-I describe various physical quantities (in SI unit) match it with List-II.

	List – I		List - II
(I)	Value of E_1	(P)	150
(II)	Value of E_2	(Q)	75
(III)	Value of $E_1 + E_2$	(R)	225
(IV)	Value of $E_1 - E_2$	(S)	175
		(T)	450
		(U)	300

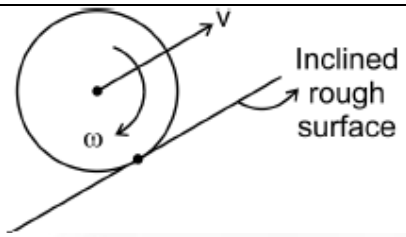
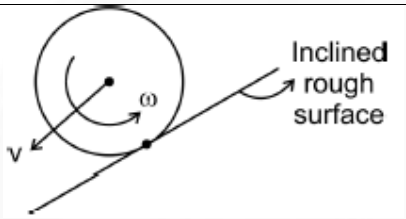
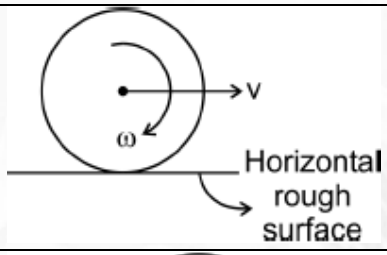
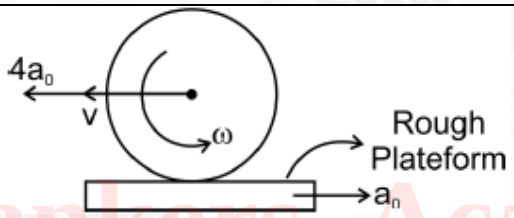
If $A_1 = 1m^2$, $A_2 = 2m^2$, $d_1 = 1\,\text{mm}, d_2 = 2\,\text{mm}$ and applied field is 100 N/C. Match List I with List II.

A) I – P , II – Q, III – R, IV – Q **B)** I – R, II – Q, III – U, IV – T

C) I – R, II – S, III – U, IV – T **C)** I – P, II – Q, III – R, IV – T

33. A round object is under pure rolling on a rough surface. Different condition of motion of round object on different rough surfaces are shown in List-I. In all cases, there is no slipping at points of contact. Direction of velocity of centre of object is to be considered as forward direction and opposite to velocity of centre, as reverse direction. Now match List-I with List-II.

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	List – I		List – II
(I)		(P)	Friction is in forward direction.
(II)		(Q)	Friction is in reverse direction.
(III)		(R)	Angular momentum of object is conserved with respect to point of contact of object.
(IV)		(S)	Friction is zero.
		(T)	Angular momentum is conserved with respect to centre of object

A) $I \to P; II \to Q; III \to S,T; IV \to Q,R$

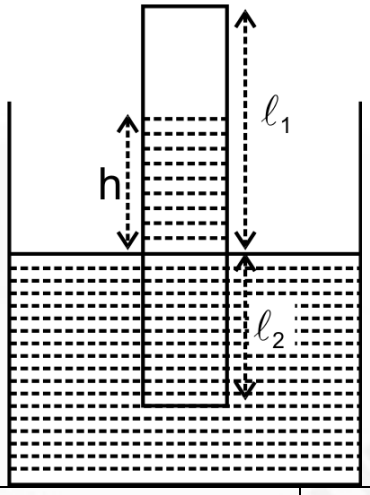
B) $I \to P,Q; II \to Q,T; III \to S,T; IV \to Q,R$

C) $I \to P; II \to Q,S; III \to R,T; IV \to Q,R$

D) $I \to P,Q; II \to Q,T; III \to R,T; IV \to Q,T$

34. A capillary is partially dipped in a liquid. Angle of contact between liquid and material of capillary is zero. Length of capillary above liquid is ℓ_1 while length of capillary inside liquid is ℓ_2 . Liquid rises upto a light h (if $h < \ell_1$) in capillary. Radius of capillary is negligible with respect to ℓ_1, ℓ_2 & h. Match List I with List II.

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	List-I		List-II
(I)	If $\ell_1 > h$ & $\ell_2 > h$ and suddenly capillary is taken out of liquid completely.	(P)	Both upper and lower free surface of liquid in capillary are hemispherical.
(II)	If $\ell_1 > h$, $\ell_2 < h$ and suddenly capillary is taken out of liquid completely.	(Q)	Length of liquid in capillary is 2h.
(III)	$\ell_1 > h$ and $\ell_2 > h$. If system (beaker + capillary) is placed inside International Space Station (ISS) where gravity is not present but atmospheric pressure is present. Now capillary is taken out of liquid (ℓ_1, ℓ_2 and h are measured when experiment is conducted on earth).	(R)	Length of liquid in capillary is $\ell_1 + \ell_2$
(IV)	$\ell_1 < h, \ell_2 < h$ and suddenly capillary is taken out of liquid	(S)	Both upper and lower free surface of liquid in capillary are flat.
		(T)	Lower free surface of liquid in capillary is not hemispherical.

A) $I \to P,Q; II \to R,T; III \to R; IV \to R,T$

B) $I \to P,R; II \to P,Q; III \to R,S; IV \to R,T$

C) $I \to P,Q; II \to T; III \to R,S; IV \to R,T$

D) $I \to P,R; II \to P,R; III \to Q,R; IV \to Q,T$

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CHEMISTRY **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

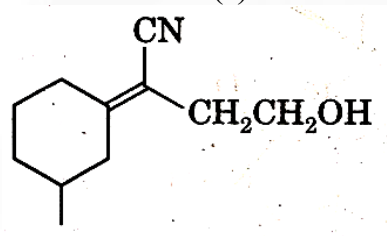
This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

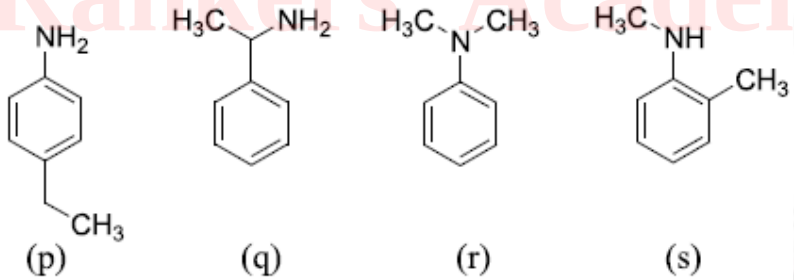
35. True statement(s) about the following compound is/are:



A) It shows geometrical isomerism **B)** It shows optical isomerism

C) Its configuration is E **D)** It is having three stereocentres

36. Given below are isomeric amines. The incorrect statement/s about them from the following is/are



A) (p) and (q) both will give unstable products respectively with $NaNO_2$ in HCl at 268 K

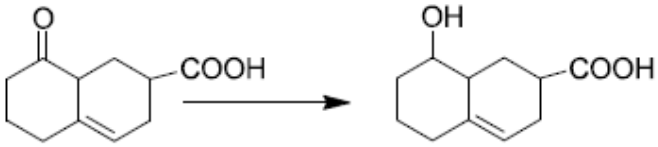
B) Reaction of all amines with HCl is exothermic

C) Reaction of benzene sulphonyl chloride with (s) gives a solid product that is soluble in NaOH

D) (q) has the highest basicity among these

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37. Which one/s of the reduction techniques mentioned below is/are NOT suitable for the following chemical transformation



A) $NaBH_4$ based reduction **B)** $LiAlH_4$ based reduction

C) DIBAL –H based reduction **D)** B_2H_6 based reduction

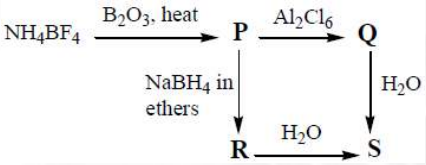
SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

38. In the following reaction sequence, identify p, q, r and s



A) $P = BF_3, Q = BCl_3, R = B_2H_6, S = B(OH)_3$

B) $P = NH_4BO_2, Q = B_2Cl_4, R = B(OH)_3, S = B(OH)_4^-$

C) $P = NH_4BO_2, Q = B(OH)_3, R = NaBO_2, S = B(OH)_3$

D) $P = BF_3, Q = B_2Cl_4, R = B(OH)_3, S = B(OH)_4^-$

39. A student mixed three sodium salts accidentally. To identify these salts, some tests were performed and found that

i. When chromyl chloride test was performed, mixture of colored gases released and NaOH solution turned yellow and gave positive test.

ii. Brown ring test was positive but brown coloured gas was not intensified on adding copper turnings on heating the mixture with conc. H_2SO_4.

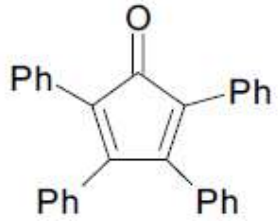
iii. Aqueous solution of the mixture was treated with $CHCl_3$ and chlorine water. From this chloroform layer $CHCl_3$ test was performed. A violet organic layer was first visible then on more addition of concentrated HNO_3, it changed to orange yellow. The observations indicate the presence of:

A) chloride, bromide and iodide **B)** nitrate, bromide and iodide

C) fluoride, nitrate and iodide **D)** chloride, nitrite and iodide

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40. The following product 'P' given below is obtained by Aldol condensation of …



Product 'P'

A) 2 moles of dibenzyl ketone, $(PhCH_2)_2CO$

B) 1 mole of dibenzyl ketone, $(PhCH_2)_2CO$ and 1 mole of acetone CH_3COCH_3

C) 1 mole of dibenzyl ketone, $(PhCH_2)_2CO$ and 1 mole of benzil (PhCOCOPh)

D) 2 moles of benzophenone, PhCOPh.

41. The results obtained when the finely divided metal powders of P, Q, R and S added to the aqueous solutions of different metal nitrates are given below:

Metal	Silver Nitrate	Lead Nitrate	Nickel Nitrate	Zinc Nitrate
P	√	√	√	X
Q	√	√	√	√
R	√	√	X	X
S	√	X	X	X

Displacement = √, No reaction = X

[Given: $E^o_{Ag^+/Ag} = 0.80V, E^o_{Zn^{2+}/Zn} = -0.76V, E^o_{Ni^{2+}/Ni} = -0.25V, E^o_{Pb^{2+}/Pb} = -0.13V$]

If these elements are to be arranged as per their positions in the electrochemical series, then which of the following will represent the above set of observations?

A) P > Q > R > S **B)** S > R > P > Q **C)** Q > P > R > S **D)** S > R > Q > P

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42. To 1 litre solution containing 0.2 M Al^{3+} and 0.2 M Zn^{2+}, NaOH is added slowly, if concentration of $[Al^{3+}]$ is 16×10^{-x} at the the time when $Zn(OH)_2$ is just begins to precipitate, then find the value of x

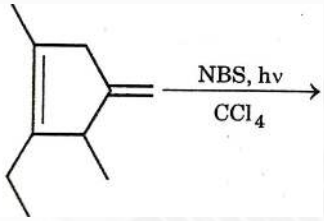
$Ksp\left[Al(OH)_3\right] = 16\times10^{-25}; K_{SP}\left[Zn(OH)_2\right] = 2\times10^{-13}$

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43. Vapour pressure of an equimolar mixture of benzene and toluene at a given temperature was found to be 80mm Hg. If vapour above the liquid phase is condensed in a beaker, vapour pressure of this condensate at the same temperature was found to be 100mm Hg. If the pure state vapour pressure of benzene and toluene is respectively x and y. Then determine value of $\frac{x+2y}{50}$

44. How many monobromo alkenes (including stereoisomers) are produced during the reaction?



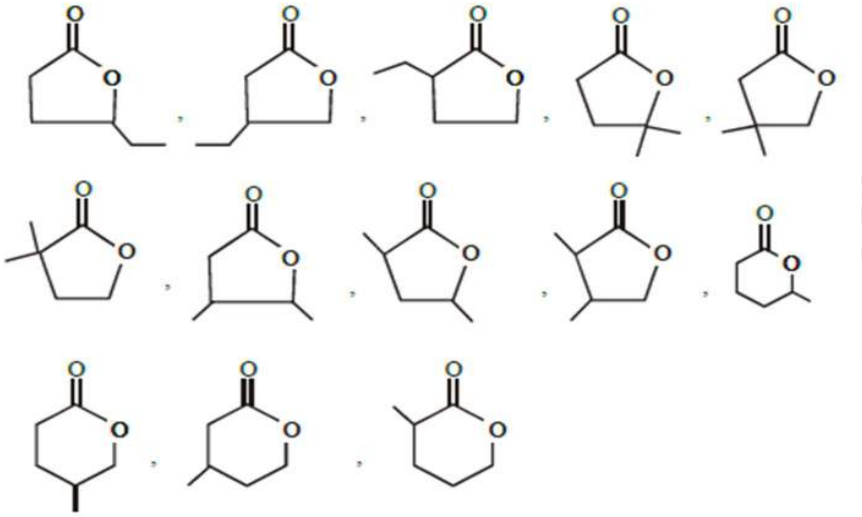
45. If electrode potential of following cell:

$$Pt_{(s)}\left|\underset{1M}{Fe^{2+}_{(aq)}},\underset{1M}{Fe^{3+}_{(aq)}}\right\|\underset{1M}{MnO_4^-(aq)},\underset{1M}{Mn^{2+}_{(aq)}},\underset{pH=5}{H^+_{(aq)}}\left|Pt_{(s)}\right.$$

is X V then calculate value of 20X.

$$\left[Given: E^o_{MnO_4^-|Mn^{2+}}=1.51V, E^o_{Fe^{3+}|Fe^{2+}}=0.78V, \frac{2.303RT}{F}=0.06\right]$$

46. How many cyclic ester can give chiral dihydroxy compounds on reaction with $LiAlH_4$ followed by $H_3O^{\oplus}$?



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47.

$$C_9H_{12}O \xrightarrow{Na} H_2 \uparrow gas$$

$$(A) \xrightarrow{CAN} Red \text{ ppt[CAN=Ceric ammonium nitrate]}$$

If A is benzenoid aromatic and insoluble in both NaOH & $NaHCO_3$ then how many chiral isomers of 'A' satisfy given conditions.

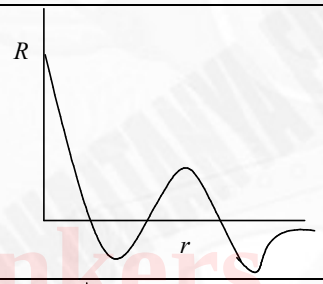
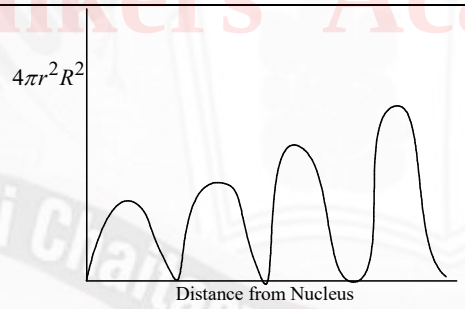
SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

48.

	COLUMN-I		COLUMN-II
A)		**P)**	4s
B)		**Q)**	$5p_y$
C)	Angular probability depends upon θ and ϕ	**R)**	3s
D)	Atleast one angular node is present	**S)**	$6d_{xy}$

A) $A \to QS; B \to PQS; C \to QRS; D \to QS$

B) $A \to P; B \to PQS; C \to QS; D \to QS$

C) $A \to PR; B \to PQS; C \to QRS; D \to QRS$

D) $A \to QS; B \to PQRS; C \to QRS; D \to QRS$

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49. Match the list – I with List –II

	List – I (Mixture)		**List – II** (Purification process)
(A)	Chloroform and aniline	**(I)**	Steam distillation
(B)	Benzoic acid and naphthalene	**(II)**	Sublimation
(C)	Water and aniline	**(III)**	Distillation
(D)	Naphthalene and sodium chloride	**(IV)**	Crystallisation

Choose the correct answer from the options given below.

A) $A \to IV; B \to III; C \to I; D \to II$ **B)** $A \to III; B \to I; C \to IV; D \to II$

C) $A \to III; B \to IV; C \to II; D \to I$ **D)** $A \to III; B \to IV; C \to I; D \to II$

50. Match The following

	Column I		**Column II**
A)	$CH_3 - CHBr - CD_3$ on treatment with alc. KOH gives $CH_2 = CH - CD_3$ as a major product	**P)**	E1 reaction
B)	$Ph - CHBr - CH_3$ reacts faster than $Ph - CHBr - CD_3$	**q)**	E2 reaction
C)	$Ph - CH_2 - CH_2Br$ on treatment with $C_2H_5OD / C_2H_5O^-$ gives $Ph - CD = CH_2$ as the major product.	**r)**	E1CB reaction
D)	$PhCH_2CH_2Br$ and $PhCD_2CH_2Br$ react with same rate.	**s)**	First order reaction

A) $A \to q; B \to q; C \to r; D \to ps$ **B)** $A \to p; B \to qr; C \to r; D \to ps$

C) $A \to q; B \to p; C \to r; D \to ps$ **D)** $A \to qs; B \to p; C \to r; D \to ps$

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51. List – I contains metal species and List – II contains their properties.

	List – I		List – II
I)	$[Cr(CN)_6]^{4-}$	**(P)**	t_{2g} orbitals contain 4 electrons
II)	$[RuCl_6]^{2-}$	**(Q)**	μ(spin-only) = 4.9 BM
III)	$[Cr(H_2O)_6]^{2+}$	**(R)**	Low spin complex ion
IV)	$[Fe(H_2O)_6]^{2+}$	**(S)**	Metal ion in +4 oxidation state
		(T)	d^4 species

[Given atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST – I with their properties in LIST – II and choose the correct option

A) $I \to PRT; II \to PSRT; III \to QT; IV \to PQ$

B) $I \to RS; II \to PTR; III \to PQ; IV \to QT$

C) $I \to PR; II \to PRST; III \to RT; IV \to PT$

D) $I \to QT; II \to PST; III \to PT; IV \to QR$

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